

**APPENDIX G — EXEMPLARY METRICS, INDICES, AND DERIVED FEATURE FAMILIES (NON-LIMITING)**

**A. Overview**

This appendix provides non-limiting examples of derived metrics, indices, and feature families that may be generated from one or more nodes. The metrics are intended to be privacy-preserving and may be presented as indices, zones, ranges, curves, distributions, and confidence indicators. Any metric described herein may be omitted, substituted, combined, renamed, or computed differently while maintaining the functional intent.

**B. Example Longitudinal External Metrics (Node 1) (Non-Limiting)**

In various embodiments, derived metrics from an external longitudinal node may include: 1. rigidity index derived from pressure/strain/displacement sensing; 2. tumescence pattern index derived from circumferential change signatures; 3. duration metrics derived from validated event windows; 4. stability metrics describing within-window variability and cross-window repeatability; 5. thermal context tags and temperature-compensated indices; 6. vascular confirmation indicators derived from PPG features; and 7. quality and confidence indicators derived from contact integrity and motion gating.

**C. Example Intraluminal Longitudinal Metrics (Node 2) (Non-Limiting)**

In various embodiments, derived metrics from an intraluminal longitudinal node may include: 1. hydration/wetness index derived from dielectric/capacitive sensing; 2. impedance-derived tissue-adjacent index derived from single or multi-frequency impedance; 3. pH trend curve and drift-compensated pH indices; 4. pelvic activity signature derived from distributed pressure sensing; 5. temperature trend curve and thermal context tags; 6. context-conditioned indices gated to stable, validated windows; and 7. quality and confidence indicators derived from contact integrity and motion gating.

**D. Example External Geometry Metrics (Node 3) (Non-Limiting)**

In various embodiments, a session-based geometry node generates: 1. length estimate and repeatability score; 2. diameter profile as a set of zone-based values or a privacy-preserving curve; 3. shape indices including curvature, tapering, and symmetry proxies; 4. scan validity label derived from scan-path constraints; 5. anti-spoof confidence derived from contact and physiologic confirmation modalities; and 6. baseline normalization parameters used to scale or interpret longitudinal metrics.

**E. Example Intraluminal Geometry/Compliance Metrics (Node 4) (Non-Limiting)**

In various embodiments, an intraluminal geometry/compliance node generates: 1. internal distance estimates and endpoint-derived profiles; 2. compliance index derived from stimulus pressure vs. response curves; 3. contact distribution indices derived from circumferential sensing elements; 4. stability and repeatability indices derived from repeated trials; 5. session validity and confidence indicators derived from motion gating and contact confirmation; and 6. baseline parameters for normalization and cross-session comparison.

**F. Example Upper-Body Context Metrics (Node 5) (Non-Limiting)**

In various embodiments, an upper-body context node generates: 1. heart rate and HRV metrics derived from ECG and/or PPG; 2. respiratory proxy metrics derived from impedance, IMU, or ECG-derived respiration; 3. activity state and posture proxy labels derived from IMU; 4. electrodermal activity indices and skin conductance indicators; 5. thermal context and temperature trends; and 6. wear integrity and contact integrity indicators.

#### **G. Cross-Device Correlation Outputs (Non-Limiting)**

In various embodiments, cross-device outputs include: 1. alignment score between selected node metrics; 2. synchrony curve computed across time-aligned windows; 3. lag signature indicating time offsets between changes; 4. context-matched correlation score computed within matched states (e.g., low-motion windows); 5. stability indicator describing repeatability across sessions; and 6. confidence indicators derived from combined quality indices.

#### **H. Cross-User Correlation Outputs (Non-Limiting)**

In various embodiments, cross-user outputs include: 1. pairwise alignment score computed from derived metrics only; 2. timing synchrony score computed from context-matched windows; 3. stability score reflecting repeatability and recency; 4. consent and scope indicators showing permission status; and 5. confidence indicators reflecting quality thresholds and validation gating.

#### **I. Variations**

Any metric family described herein may be implemented using rules-based processing, statistical models, and/or machine learning models. Metrics may be computed locally or remotely, and may be normalized to within-user baselines and/or population-referenced distributions. Outputs may remain privacy-preserving and may avoid explicit anatomical imagery and explicit anatomical labeling.